

Royal Netherlands Institute for Sea Research

FAIR - UNECSS workshop

Best Practices for Making (Meta)Data FAIR

Antsje van der Leij & Flavio Francisco

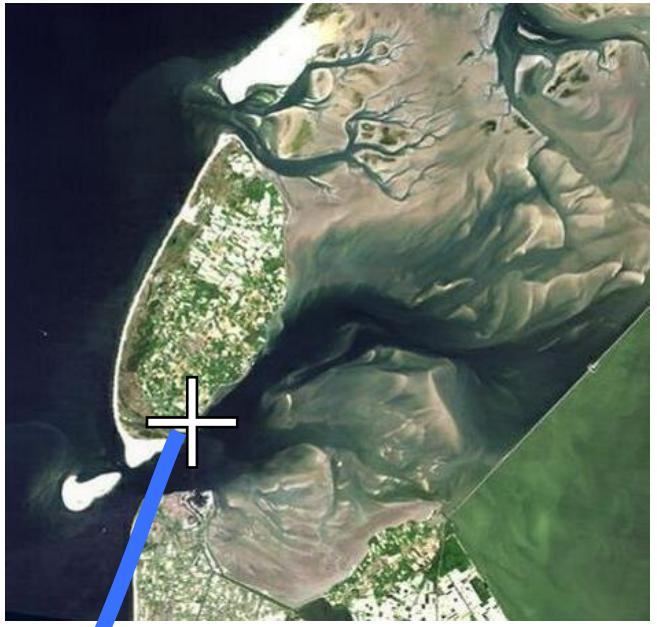
Alkmaar - Netherlands

Royal NIOZ at a glance

- On the island of Texel and Yerseke
- Major working areas:
- North/Mid Atlantic, North Sea, Mediterranean, etc
- Staff: 200 people + 200 flex
- Founded in 1876



150 YEARS ROYAL NIOZ | SEA SCIENCE SOCIETY





RV Anna Weber-van Bosse





NIOZ fleet

RV Anna Weber-Van Bosse

RV Wim Wolff

RV Adriaen Coene

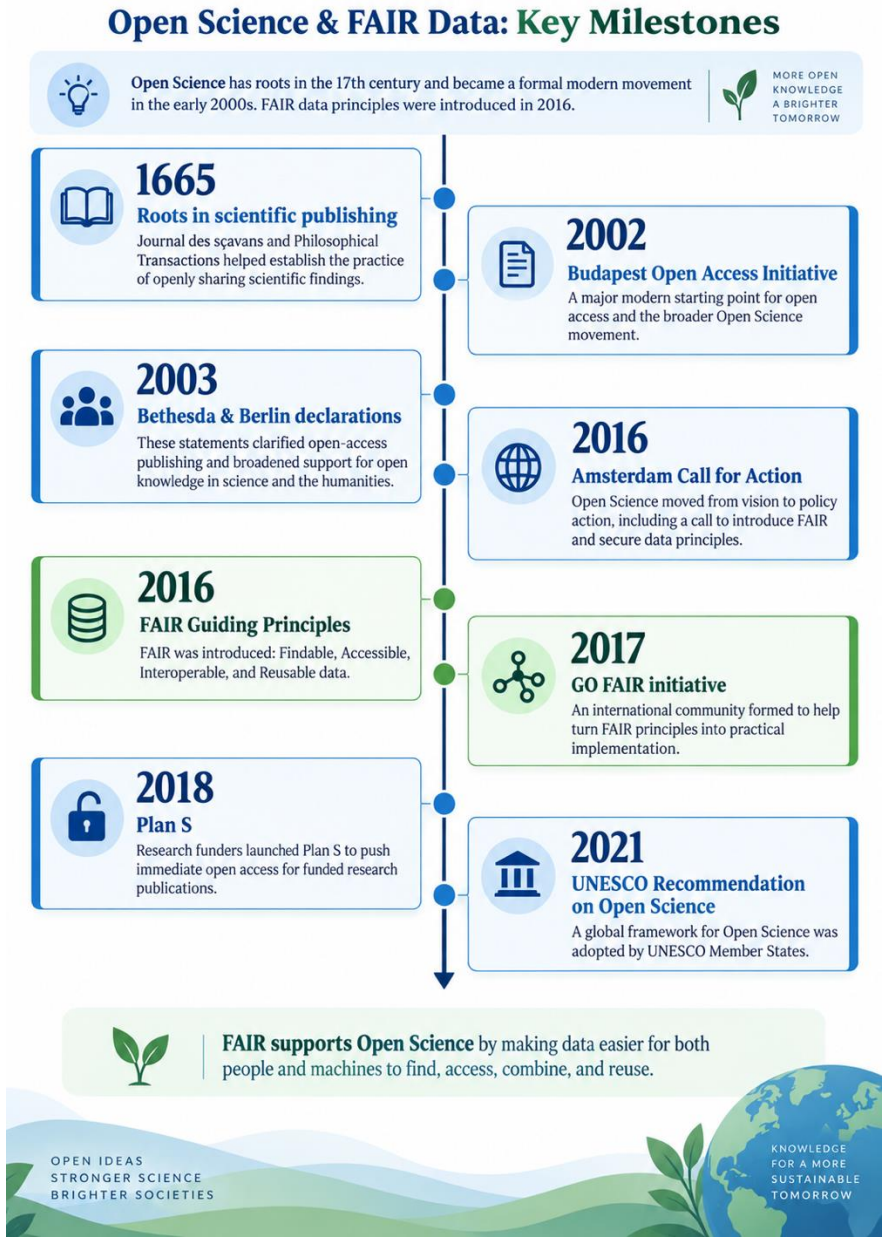



F
Findable
A
Accessible
I
Interoperable
R
Reusable

Key Milestones

Open Science & FAIR Data

- **Open Science** makes research more transparent, accessible, reproducible, and collaborative.
- **2002–2003:** Major Open Access initiatives, including the **Budapest Open Access Initiative** and **Berlin Declaration**, accelerated the movement.
- **2016:** The **FAIR Guiding Principles** were formally published.
- **FAIR = Findable, Accessible, Interoperable, Reusable.**
- FAIR data supports Open Science by making research outputs easier for **people and machines** to discover, understand, share, and reuse.



What does FINDABLE mean?

- F1. (Meta)data are assigned a globally unique and persistent identifier
- F2. Data are described with rich metadata
- F3. Metadata clearly and explicitly include the identifier of the data they describe
- F4. (Meta)data are registered or indexed in a searchable resource

Wilkinson, M., Dumontier, M., Aalbersberg, I. *et al.* The FAIR Guiding Principles for scientific data management and stewardship. *Sci Data* **3**, 160018 (2016). https://doi.org/10.1038/sdata.2016.18

What does FINDABLE mean?

F1. (Meta)data are assigned a globally unique and persistent identifier

Conditions for your identifier:

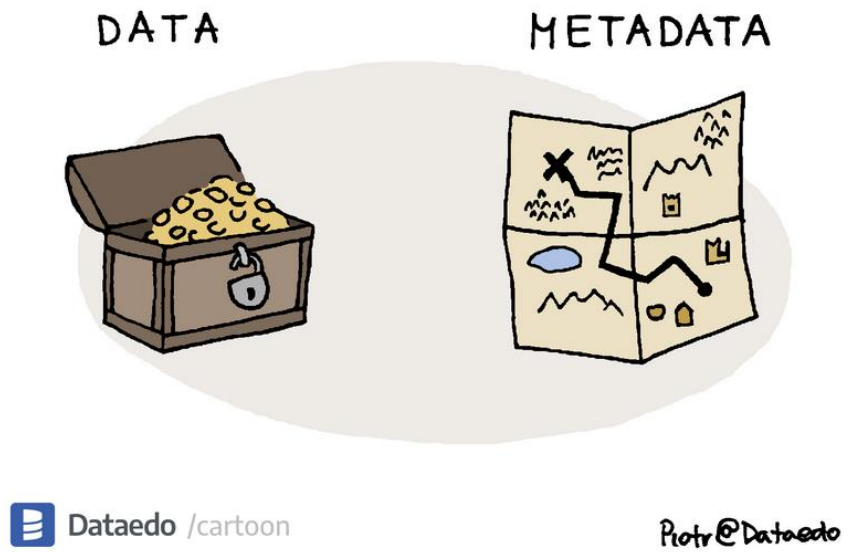
- **Globally unique:**
  Someone else could not reuse/reassign the same identifier without referring to your data.
  Globally unique identifiers can be obtained from a registry service that uses algorithms guaranteeing the uniqueness of newly minted identifiers.

- **Persistent:**
  It takes time and money to keep web links active, so links tend to become invalid over time.
  Registry services guarantee resolvability of the link into the future, at least to some degree.

- **Examples:**
  DOI: https://doi.org/10.25850/nioz/7b.b.ud
  ORCID: https://orcid.org/0009-0008-7312-1512

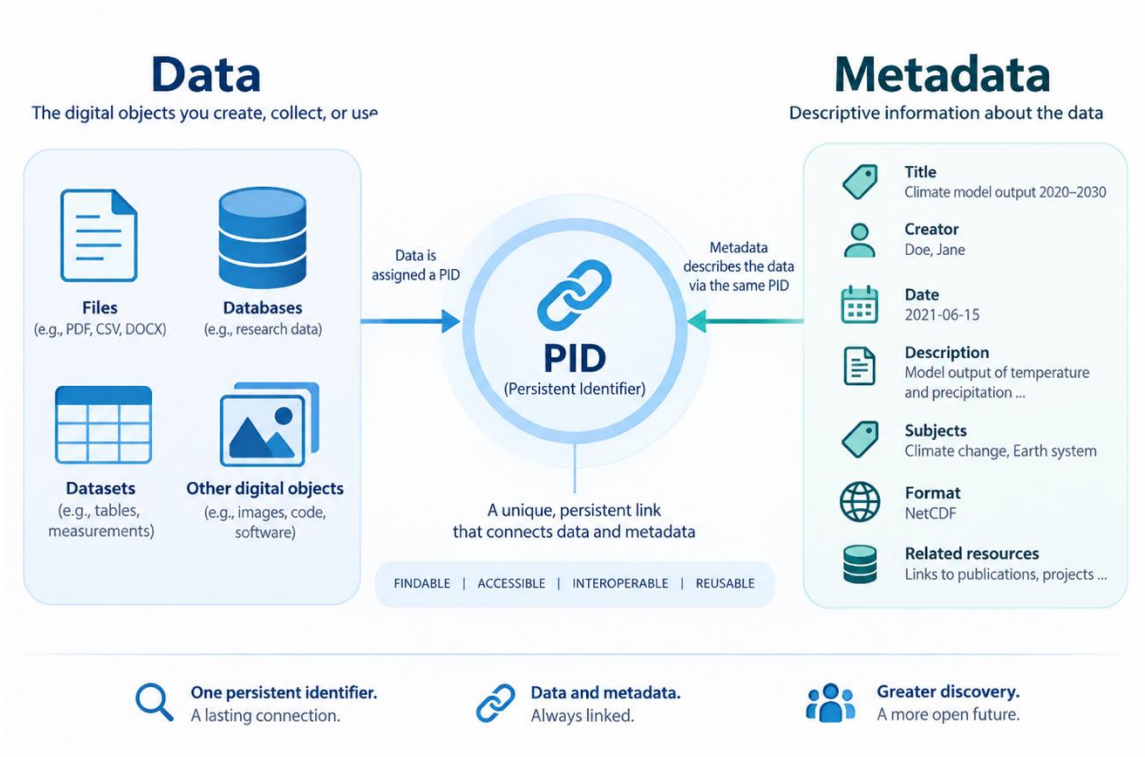
What does FINDABLE mean?

F2. Data are described with rich metadata



What does FINDABLE mean?

F3. Metadata clearly and explicitly include the identifier of the data they describe



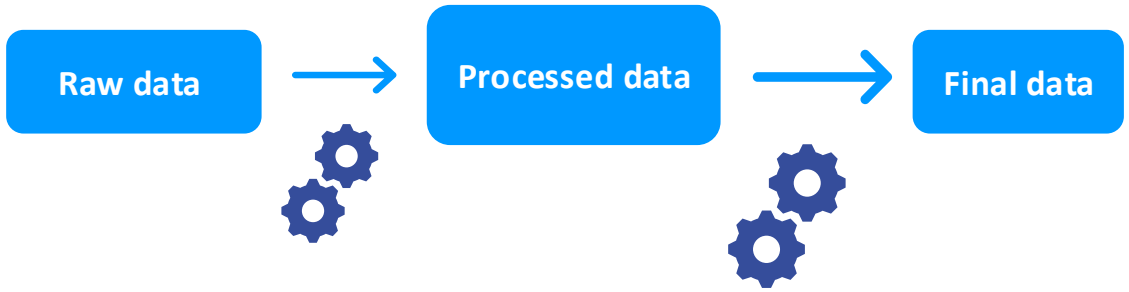
What does FINDABLE mean?

F4. (Meta)data are registered or indexed in a searchable resource



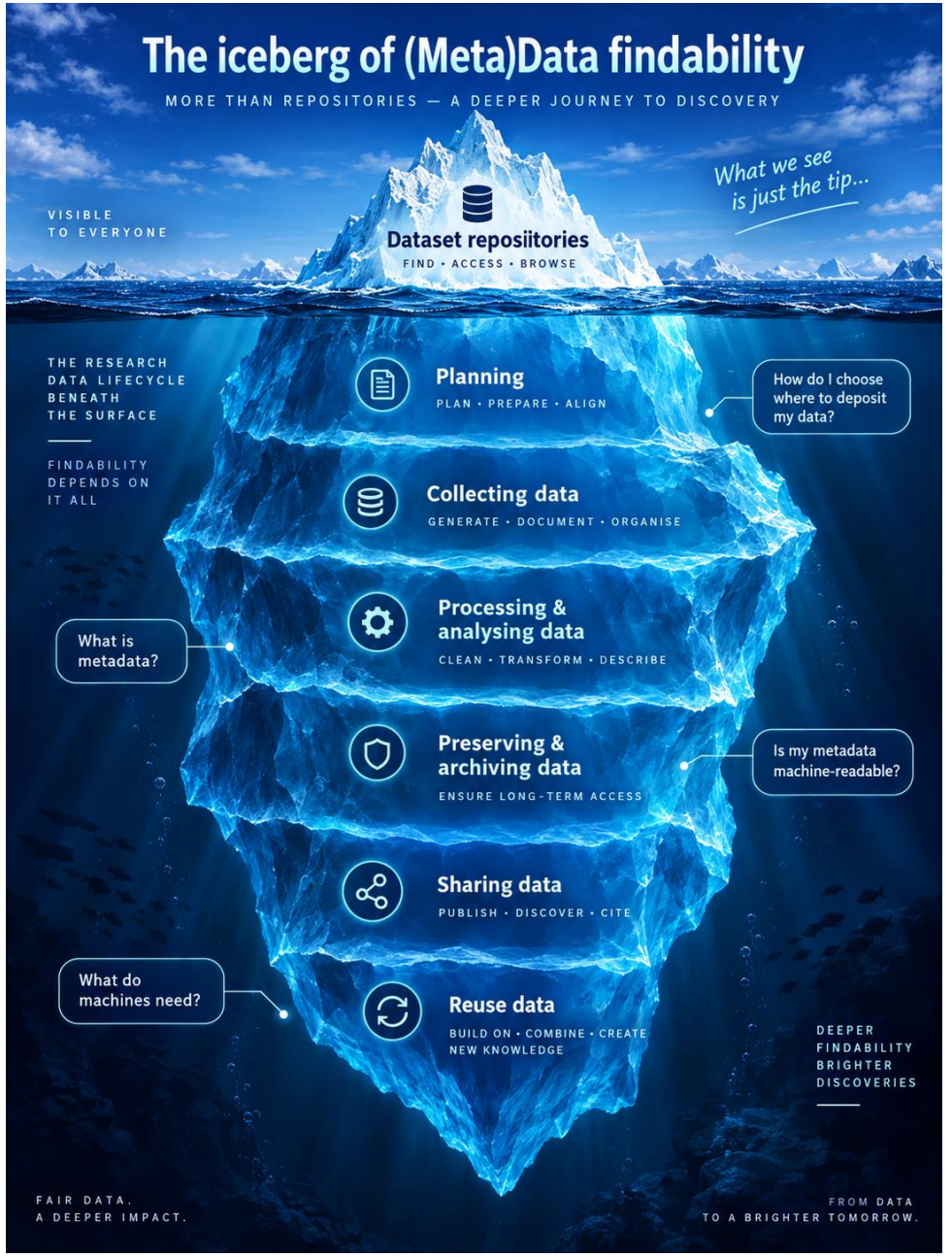
The iceberg of (Meta)data findability

Data goes through steps during the research data lifecycle:



Each dataset needs to be published & perhaps not in the same place

- Metadata can be different for each dataset
- Each dataset can have different purpose



The iceberg of (Meta)data findability

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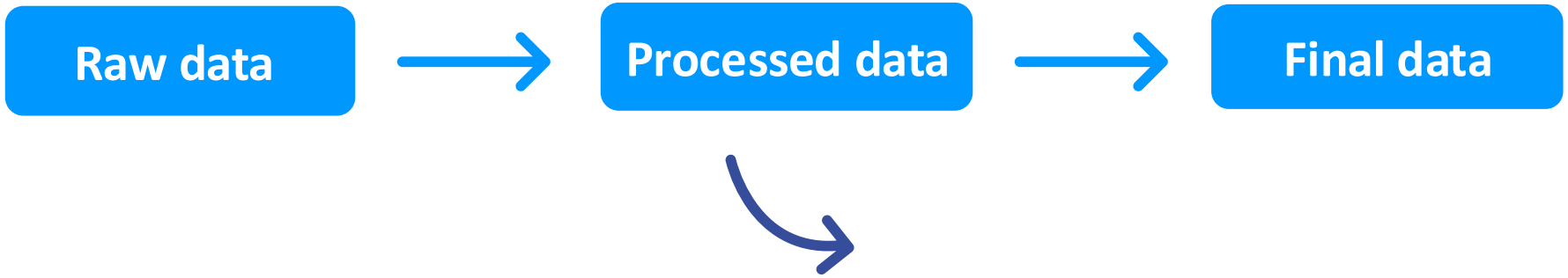


Raw data are the foundation of your science

They cannot be recreated without repeating the experiment

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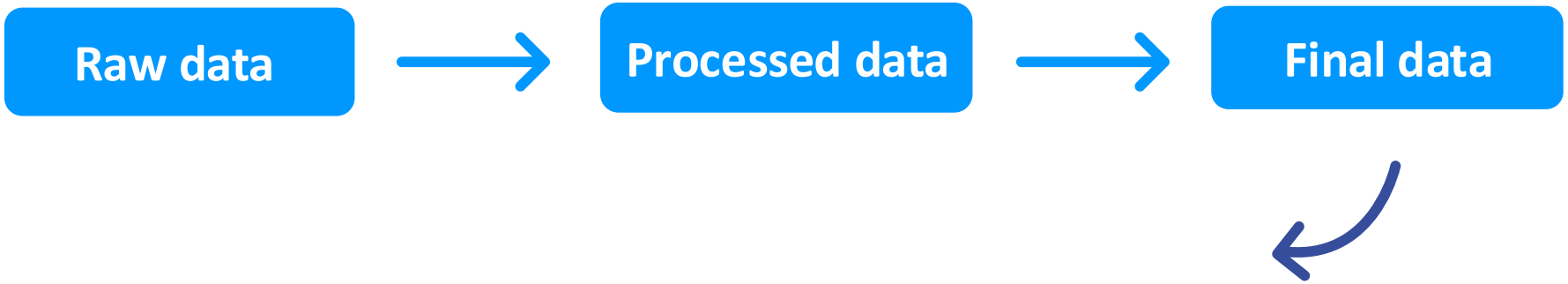


Processed data are the outputs of data analysis and useful for trustability and reusability.

If you can publish data analysis scripts + raw data, you do not necessarily need to also publish the processed data

The iceberg of (Meta)data findability

Data goes through steps during the research data lifecycle:



- Final data are the immediate evidence of your science
- They are often part of your paper
- They should also be published if they are different to the raw or processed data

Sharing your data

There are many ways to publish research data & make them findable:

- Publish your research data on a project website or as supplementary information to your research paper.

  Not enough to make your data truly findable.

- Make research data publicly available by publishing it in a public **Data repository** together with its **metadata**.

  The most recommended & FAIR way

  - Your data are preserved & are accessible for the long term
  - Your data will be (more) standardised
  - Your data can be searched for by the scientific community; a repository is a go to find data
  - Your data will have a license, which makes the data rights explicit & facilitates correct re-use of your data.

- Publish your data in a data paper

  - In case your data constitutes a large dataset, which can be used on its own for future research
  - Peer-reviewed describing openly accessible data sets for future reuse

  But still publish the data
  And add the link to it!

Data repositories

A database infrastructure that collects, manages and stores data for preservation & and sharing.

TYPES of repositories:

Generic repository

Which allows to deposit data independent to the scientific domain or data type

zenodo

MARINE DATA ARCHIVE

Domain-specific repository

Which is narrower in scope & only manages specific data types. It depends on your field whether a suitable repository already exists

Pangaea

EMODnet European Marine Observation and Data Network

GBIF

ENA European Nucleotide Archive

EurOBIS European Ocean Biodiversity Information System

Institutional repository

It holds data from a particular research institution

Universities, Institutes, etc.

How to choose where to deposit your data?

Research domain's consensus

Researchers' best practices

Journal recommendations

Data repository registries

re3data.org
REGISTRY OF RESEARCH DATA REPOSITORIES

FAIRsharing.org
standards, databases, policies

How to choose where to deposit your data?

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Research domain's consensus

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Journal recommendations

Data repository registries

Trustworthy repository →

- They have received certification: e.g., Core Trust Seal
- Domain-specific repositories that are internationally recognized, commonly used & endorsed by community
- Or FAIR-enabling repositories: (generic) repositories with essential characteristics such as **authentication & authorization, machina-actionable metadata, long-term preservation**

How to choose where to deposit your data?

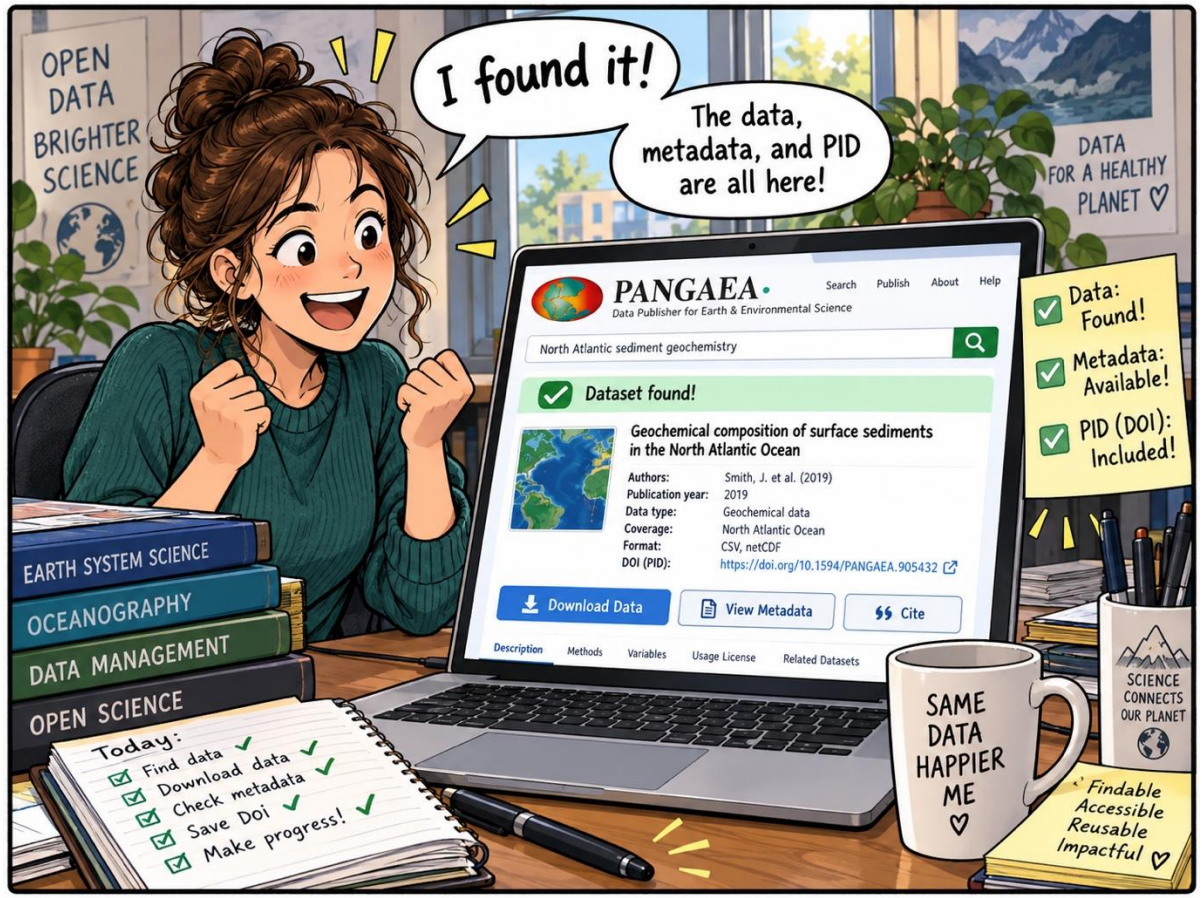
Keep the following basic requirements in mind when choosing a repository:

- A persistent identifier should be assigned to each dataset
- Rich and standardized metadata should be present
- Terms of use information (license) – CC BY 4.0
- Open infrastructure to access content on the platform
- Preservation & continued availability



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PANGAEA.
Data Publisher for Earth & Environmental Science

GBIF 25

OBIS OCEAN BIODIVERSITY INFORMATION SYSTEM

ENA European Nucleotide Archive

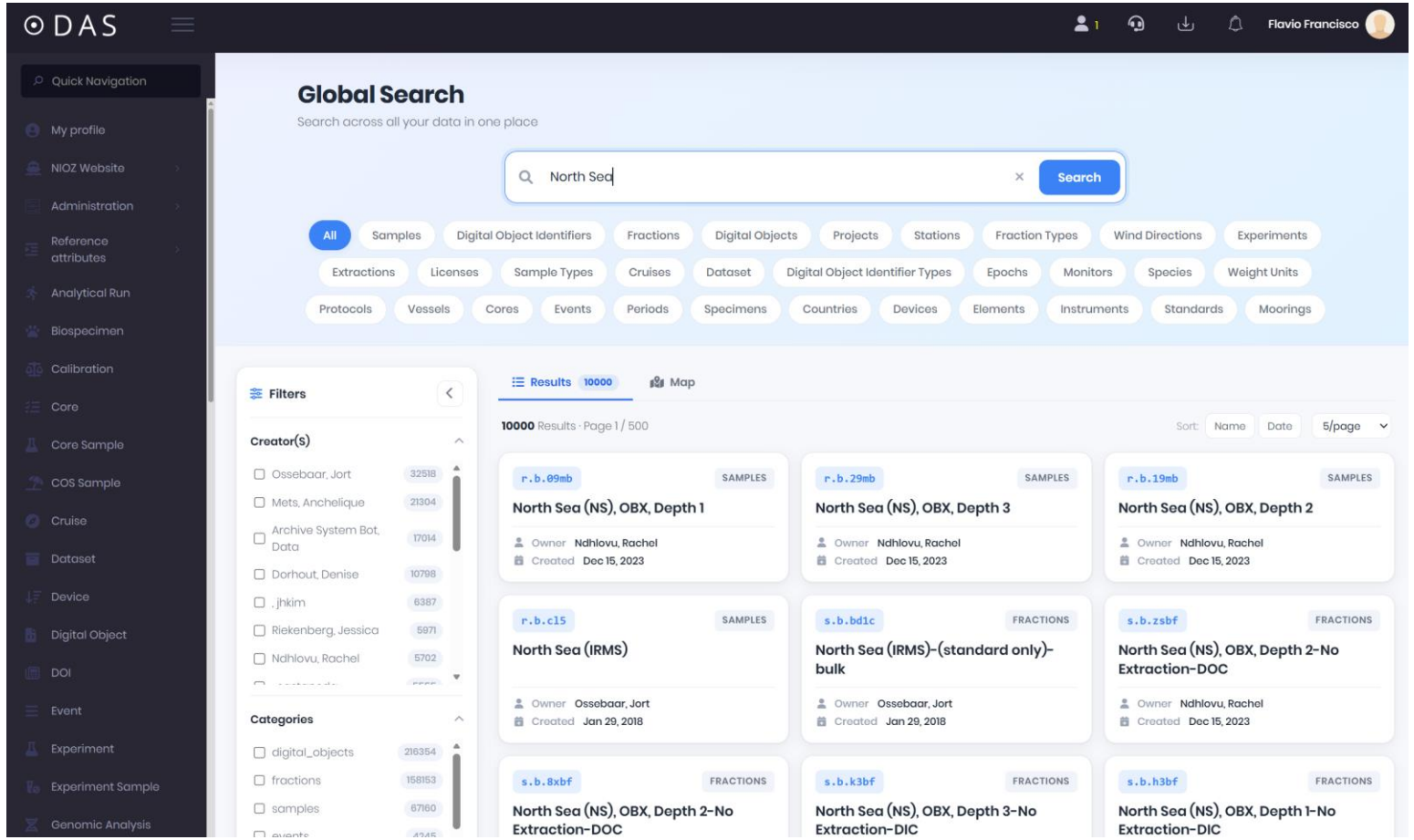
zenodo

Open Research Europe

EMODnet

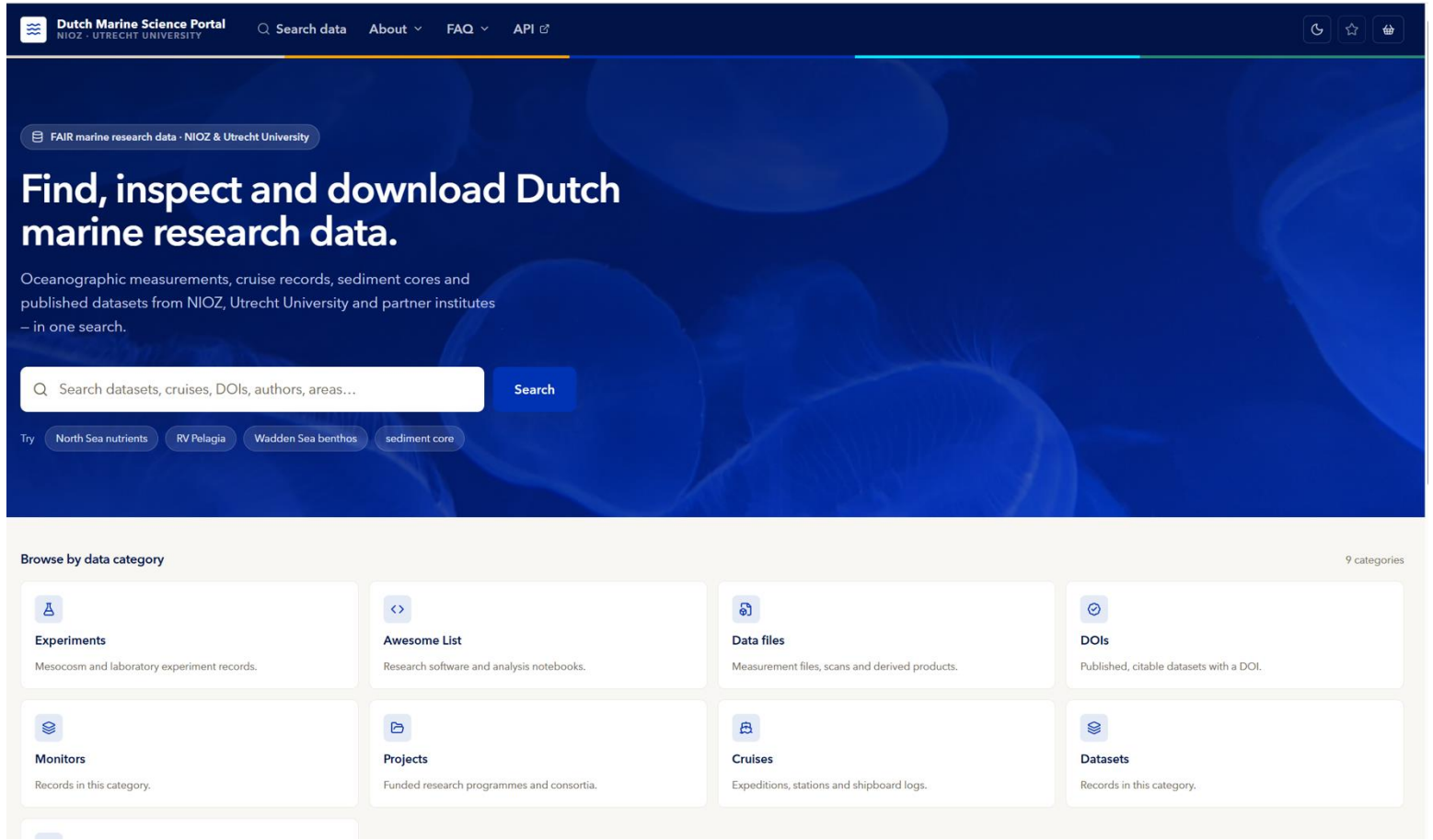
What does FINDABLE mean?

Data Archive System (DAS) – https://das.nioz.nl



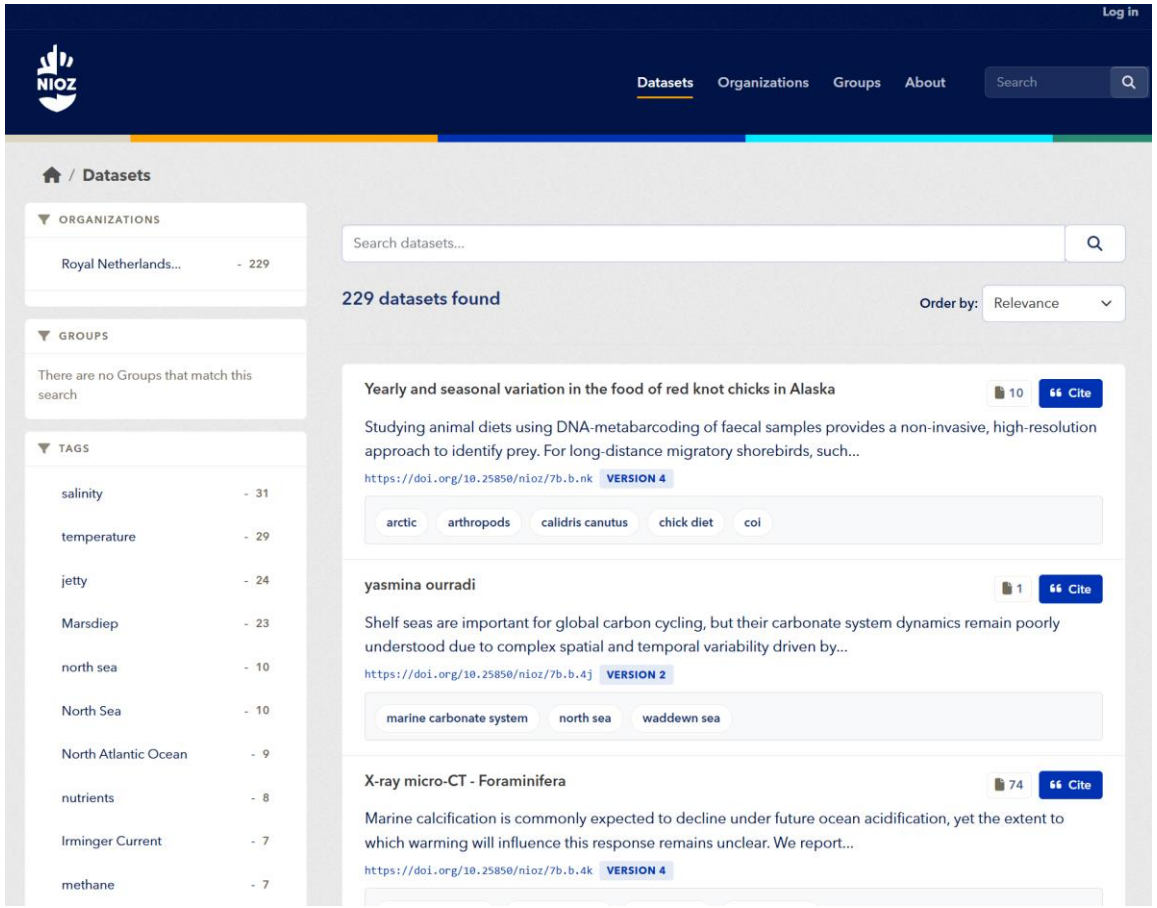
What does FINDABLE mean?

Dutch Marine Science Portal – https://scienceportal.nioz.nl



What does FINDABLE mean?

NIOZ Dataset repository – https://doi.nioz.nl



QUESTIONS AND HANDS ON TIME

